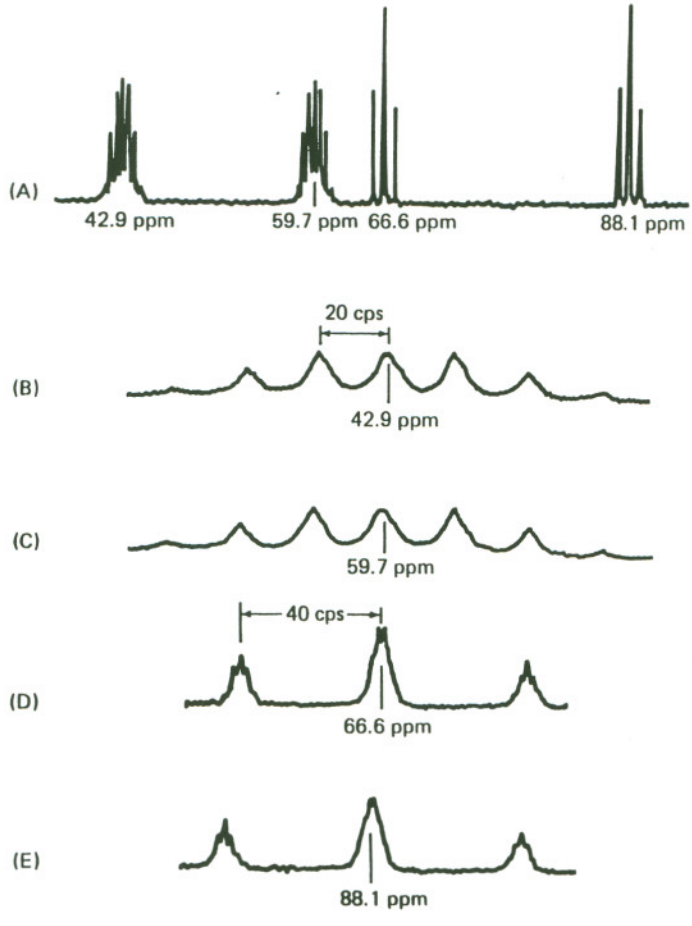
INORGANIC CUMULATIVE EXAM

February 11, 2000

The following problems were copied from a popular textbook on the application of physical methods in inorganic chemistry. There are 3 separate problems. Provide the answers to each.

1. Which complex would have more N-O vibration: $(NH_3)_5CoNO_2^{2+}$ or $(NH_3)_5CoONO^{2+}$?

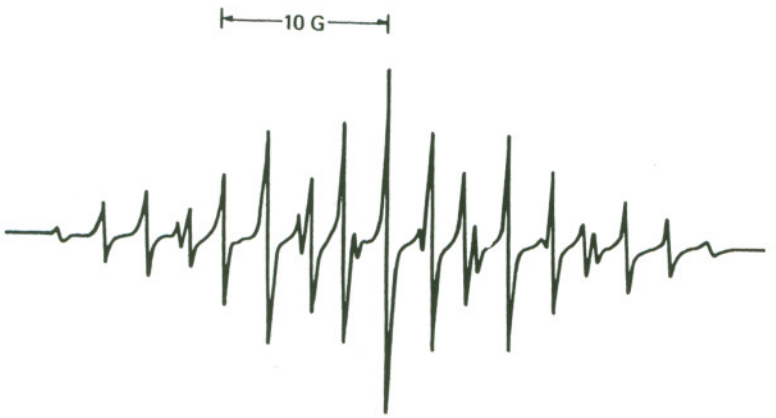
2. The ^{19}F nmr spectrum of $PF_3(NH_2)_2$ is reported in (a), and expanded spectra of the individual peaks are presented in (b) through (e). Propose a structure and interpret the spectrum.



P and F have a nuclear spin of ½

SEE NEXT PAGE

3. The epr spectrum of the pyrazine anion, , is given as follows. Interpret the spectrum and calculate the *a*-value(s).



N has a nuclear spin of 1